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Received: 29 August 2020 | Accepted: 22 June 2021

DOI: 10.1111/ppa.13431

Plant Pathology WILEY

ORIGINAL ARTICLE

First report of oospore formation in *Plasmopara viticola*, the causal agent of grapevine downy mildew, in highland regions of southern Brazil

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Funding information
UNIEDU/FUMDES

Abstract

This work evaluates the formation of oospores of *Plasmopara viticola*, the causal agent of grape downy mildew (DM), in highland regions in southern Brazil. Leaves of susceptible and resistant grape genotypes naturally infected with the pathogen were collected in the autumn of 2017, 2018, and 2019 from vineyards located in the highlands of Santa Catarina state. Leaf tissues were evaluated by light microscopy and scanning electron microscopy. Oospores of *P. viticola* were identified in both susceptible and resistant host genotypes. They were concentrated in the central regions of the DM lesions, close to the leaf veins, and exhibited a rounded shape, yellowish colour, thick wall, and a diameter ranging from 16.28 to 49.15 μm. The formation of oospores is strong evidence that sexual reproduction is occurring in *P. viticola* in the climatic conditions of the highlands of southern Brazil. Sexual reproduction contributes to the maximization of genetic diversity via meiosis. Populations with high genetic variability are more likely to break resistance mechanisms conferred by resistance genes and to develop resistance to fungicides applied for disease control. To our knowledge, this is the first scientific study to prove the formation of *P. viticola* oospores in Brazil. The results presented provide a solid basis for further studies on sexual recombination in *P. viticola*. Genetic improvement programmes for grapevines, disease management methods, and disease prediction models need to consider the sexual reproduction of this pathogen, otherwise their effectiveness may be compromised.

KEYWORDS
oomycete, oospore, *Plasmopara viticola*, sexual reproduction, *Vitis* spp.

1 | INTRODUCTION

In the last two decades, highland regions of Santa Catarina, southern Brazil, with altitudes ranging from 900 to 1400 m above sea level, have become areas for the cultivation of European grapevine varieties (*Vitis vinifera*) and the production of high quality wines (Malinovski et al., 2012). At these higher locations, the climate is colder, leading the plants to accumulate degree days at a slower rate, increasing the length of the annual grapevine cycle; this allows the grapes to ripen during a period of cooler nights, improving the wine quality (Falcão et al., 2010). The wines produced in these regions have a high phenolic compound content, in accordance with or superior to those reported in the literature from the most renowned regions in the world (Gris et al., 2011). For example, this region has

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demonstrated high potential for production of red wines rich in stilbenes and tyrosol, with concentrations of total stilbenes (sum of *cis*- and *trans*-piceid and *cis*- and *trans*-resveratrol forms) ranging from 12.7 to 52.8 mg/L. These values are higher than those reported by Goldberg et al. (1995) for 14 regions worldwide. These wines also demonstrated significant in vivo antioxidant activity.

However, the high level of precipitation during the growing season (around 1200 mm) favours the development of downy mildew (DM), and the European varieties are highly susceptible to its causal agent. Prophylactic applications of fungicides are required to avoid damage caused by the disease, which increases production costs and can have negative impacts on the environment and human health (Taylor & Cook, 2018).

Downy mildew is a disease of great importance worldwide, including the southern region of Brazil, and its causal agent is *Plasmopara viticola* (Gessler et al., 2011). The reproductive cycle of the pathogen is divided into the asexual phase, with the formation of zoospores, and the sexual phase, with the formation of oospores (Burruano, 2000). This oomycete is heterothallic, requiring the interaction of two compatible mating types for sexual reproduction (mating types P1 and P2; Wong et al., 2001). The mating type loci have been recently identified and this 570-kb repeat-rich nonrecombining region showed two highly divergent alleles (Dussert et al., 2020).

The formation of oospores begins when the antheridium encounters a compatible oogonium wall (Vercesi et al., 1999). After contact, the fertilization tube of the antheridium breaks the oogonium wall and deposits a nucleus that will fuse with a nucleus of the oogonium. The nuclei that do not fuse with the nucleus from the antheridium migrate to the periplasmic space and degenerate. Afterwards, a plasma membrane is formed that delimits the periplasm of the oosphere. After the formation of the oosphere, the composition of the cytoplasm changes, with vacuoles and lipid bodies randomly distributed and found in large quantities. The process of formation of the oospores ends with the reorganization of the reserve substances, where the vacuoles gradually come together to form the ooplast, and simultaneously, the lipid globule layer is formed (Burruano, 2000; Dick, 1995; Vercesi et al., 1999). In autumn, the leaves fall to the ground where the oospores overwinter. Under favourable conditions the oospores begin to germinate in the spring producing macrosporangia that contain zoospores. Through rainfall and wind, the primary zoosporangia (macrosporangia) are released and dispersed in the environment, invading the vegetative parts of the vine to cause primary infections (Gessler et al., 2011). Several asexual infection cycles occur during the vegetative cycle of the grapevine.

The presence of oospores is strong evidence that sexual reproduction is taking place in the pathogen population, contributing to increased genetic diversity of the pathogen via genetic recombination (Gobbin et al., 2006). Pathogen genetic diversity can have a direct impact on resistance to fungicides and the durability of host resistance genes. Grapevine breeding programmes introgressed resistance genes (*Rpv1*, *Rpv3*, *Rpv10*, and *Rpv12*) into elite *V. vinifera* germplasm, and in the last three decades various DM resistant varieties have been developed. These varieties are known as PIWI, an acronym from the German word *Pilzwiderstand*, which means fungus resistance. These PIWI varieties have different *Rpv* (Resistance to *P. viticola*) major resistance genes, and manifest different levels of resistance to *P. viticola* in the field (Zanghelini et al., 2019). A few of them possess two *Rpv genes* and manifest higher resistance to *P. viticola*.

Taking into consideration that resistance genes are a limited resource and their introgression into elite varieties is costly and time demanding, the preservation of their efficacy over time is a tremendous challenge (Bourguet et al., 2016). Durability is dependent on the intrinsic properties of the pathosystem and on external factors (Brown, 2015). The biology of the pathogen, and in particular the mode of reproduction, is an intrinsic factor that plays an important role, as it affects genetic diversity within and between individuals of the same population (Bourguet et al., 2016). Populations with high genetic diversity represent a greater risk of evolution as new genotypes can be selected in response to different environments, resistance genes, or fungicides. The fittest genotypes (allele combinations) can be then maintained through asexual reproduction and thus increase in frequency (McDonald & Linde, 2002). In perennial fruit crops such as grapevine, durability of host resistance is especially important due to the longevity of the orchards. The whole procedure of breeding a new cultivar by traditional methods, starting from the initial cross up until the release of a cultivar, lasts about 25 years (Eibach & Töpfer, 2015).

Although *P. viticola* was first described in Brazil in 1891 (Inglez de Souza, 1996), almost nothing is known about the biology of the pathogen under Brazilian climatic conditions. Investigations in Brazil are related to disease management (Pinto et al., 2012), pathogen incidence and severity (Zanghelini et al., 2019), and the identification of resistant host genotypes (Sanchez-Mora et al., 2017). A single study cited ongoing research into the pathogen's sexual cycle but results of this research were not published in the scientific literature (Cirne et al., 1988).

Investigations of the population genetics of the pathogen have only recently been carried out. In the south-eastern region of Brazil (State of São Paulo) only a single cryptic species of *P. viticola* (clade *aestivalis*) was found and the authors concluded that pathogen reproduction is carried out by zoospore production and that sexual reproduction had little influence on the pathogen's life cycle (Camargo et al., 2019). It is worth mentioning that the climatic conditions in south-eastern Brazil are favourable for grapevine growth all-year-round, favouring asexual survival of the pathogen. On the other hand, greater genetic diversity was observed in populations of *P. viticola* from southern Brazil compared to populations from south-eastern Brazil (Santos et al., 2020). The former showed a greater number of distinct multilocus genotypes (MLGs), lower clonal fraction (CF), and random mating, suggesting the occurrence of sexual reproduction under these environmental conditions. Southern Brazil has colder autumns and winters, especially in high altitude regions, inducing grapevine bud dormancy and complete leaf fall. In this region, the temperatures during the autumn correspond to the temperatures reported as favourable for oospore production (Grünzel, 1961; Lehoczky, 1956).

Therefore, the main hypothesis of the present investigation was that sexual reproduction may occur in the highland region of Santa Catarina State, southern Brazil, and, consequently, oospores could be

found in leaf tissues. Leaves showing typical symptoms of grapevine DM were collected in autumn from a susceptible host in two locations of the highland region of Santa Catarina and evaluated for the presence of oospores through histopathological analysis. After the confirmation of oospore formation in the susceptible host, we also evaluated if the sexual structure was formed in PIWI varieties.

2 | MATERIALS AND METHODS

2.1 | Climatic data from collection sites

Temperature (°C) and rainfall (mm) data were collected from meteorological stations located near the vineyards where DM samples were collected: Curitibanos—A860, located 4 km away from the commercial vineyard; UFSC Curitibanos—1026, located in the Experimental Agricultural Station of UFSC, and São Joaquim (SJ)—A815, located inside the Epagri Experimental Station of São Joaquim (EESJ). As temperature under 14 °C is an indicator for DM oospore formation (Gehmann et al., 1987), the temperature data from January to April were collected from these three meteorological stations to calculate the growing degree-days (GDD) using the equation GDD = $[(T_M + T_m)/2] - T_b$ (Arnold, 1960), where: T_M = maximum daily temperature (°C); T_m = minimum daily temperature (°C); and T_b = minimum threshold temperature (14 °C).

2.2 | Sample collection and conservation

Twenty leaves from the DM susceptible variety Cabernet Sauvignon (*V. vinifera*) were collected in 2017 from a commercial vineyard located in the municipality of Curitibanos (CUR1; 27°16′15.7″S, 50°37′54.5″W, 987 m a.s.l.), and in 2018 from an experimental vineyard located at EESJ, SJ (28°16′28.0″S, 49°55′53.1″W, 1,415 m a.s.l.), in the State of Santa Catarina, southern Brazil. In 2019, 10 leaves from each of 10 PIWI varieties carrying different *Rpv* loci and from variety Isabella (*Vitis labrusca* interspecific hybrid) were also collected from the experimental vineyard located at the Agricultural Experimental Station of the Federal University of Santa Catarina, Campus of Curitibanos (CUR2; 27°16′58″S, 50°35′04″W, 1,000 m a.s.l.). All leaves were sampled in the autumn (April), at the beginning of leaf senescence. The collected leaves were naturally infected by the pathogen and showed a typical mosaic symptom, as described by Vercesi et al. (1999). Information about the hosts from which leaves were collected is summarized in Table 1.

To preserve the samples, the leaves (in total 150 leaves) were fixed in a solution containing 2.5% glutaraldehyde and 0.1 M phosphate buffer (pH 7.2) for 12 hr. The samples were then dehydrated in ethanol solutions in ascending concentration (20%, 30%, 40%, 50%, 60%, and 70%), cleared using sodium hypochlorite (1%; 1:1) for 5 min (Lehoczky, 1956) and stored in 70% ethanol at room temperature until further analyses.

2.3 | Identification and morphological characterization of *P. viticola* oospores

Twenty leaves of Cabernet Sauvignon from both Curitibanos (collected in 2017) and SJ (collected in 2018), previously fixed and cleared, were selected for oospore identification and morphological characterization. From each leaf, three discs of 12 mm diameter

TABLE 1 Dates of sampling, varieties, parentage, breeder, year of crossing, loci of resistance to downy mildew, and location of grapevines evaluated in highland regions of southern Brazil

Collection date	Variety	Parentage	Breeder	Year of crossing	Loci of resistance	Location
2017	Cabernet Sauvignon	Cabernet Franc × Sauvignon Blanc	Spontaneous cross	Unknown	None	CUR1
2018	Cabernet Sauvignon	Cabernet Franc × Sauvignon Blanc	Spontaneous cross	Unknown	None	SJ
2019	Isabella	Labrusca × Meslier Petit	J. K. Stayman	Unknown	Unknown	CUR2
	Calandro	Domina × Regent	R. Eibach, R. Töpfer	1984	*Rpv3.1*	CUR2
	Cabernet Cortis	Cabernet Sauvignon × Solaris	N. Becker	1982	*Rpv10*, *Rpv3.3*	CUR2
	Baron	Cabernet Sauvignon × Brönner	N. Becker	1983	*Rpv10*, *Rpv3.3*	CUR2
	Calardis Blanc	Calardis Musque × Seyve Villard 39-369	R. Eibach, R. Töpfer	1993	*Rpv3.1*, *Rpv3.2*	CUR2
	Prior	Freiburg 4-61 × Freiburg 236-75	N. Becker	1987	*Rpv3.1*, *Rpv3.3*	CUR2
	Bronner	Merzling × Geisenheim 6494	N. Becker	1975	*Rpv10*, *Rpv3.3*	CUR2
	Felicia	Sirius × Vidal Blanc	R. Eibach, R. Töpfer	1984	*Rpv3.1*, *Rpv3.3*	CUR2
	Regent	Dian × Chambourcin	G. Alleweldt	1967	*Rpv3.1*	CUR2
	Gf.2004–043-0015	Advanced breeding line	R. Eibach	2004	*Rpv1*, *Rpv3.1*	CUR2
	Gf.2004–043-0024	Advanced breeding line	R. Eibach	2004	*Rpv1*, *Rpv3.1*	CUR2

Note: The information was obtained from the Vitis International Variety Catalogue—VIVC (Maul et al., 2020).

were excised from regions displaying typical lesions of DM, totalling 60 leaf discs per location. The leaf discs were mounted on a glass microscope slide in distilled water and the oospores were visualized in bright field using an Olympus BX-60 microscope (200×). Images of leaf disc regions containing oospores were captured with an Olympus DP73 digital camera and the diameters of 10 oospores/disc were measured randomly using the Olympus cellSens software, totalling 600 oospores per location.

2.4 | Distribution of *P. viticola* oospores inside the leaf tissue

Four leaf discs of Cabernet Sauvignon rich in oospores were selected from both locations. The leaf discs were dehydrated in ascending ethanol concentration up to 96%. The samples were embedded for 12 hr (overnight) in a pre-infiltration solution 1:1 (ethanol 96%: infiltration solution, 50 ml hydroxyethyl methacrylate and 0.5 g benzol peroxide). The samples remained in the infiltration solution for 3 hr and were embedded in Historesin (15 ml infiltration solution and 1 ml dimethyl sulphoxide). Histological sections of 5 μm were prepared using a CUT 4055 rotating microtome (microTec). The sections were stained separately with toluidine blue (O'Brien et al., 1964) and eosin Y (Kortekamp, 2005). The images were captured in an Olympus BX-60 microscope coupled with an Olympus DP73 digital camera and analysed using the Olympus cellSens software.

2.5 | Scanning electron microscope analysis

Leaf discs rich in oospores were selected from the variety Cabernet Sauvignon, immersed in liquid nitrogen in a porcelain crucible and broken into small pieces with tweezers (adapted from Kortekamp et al., 1998). The broken leaf segments were then dehydrated in ascending ethanol concentration (80%, 90%, 100%), performed in three replicates of 10 min per concentration, followed by dehydration using an EM-CPD-030 critical point dryer (Leica). Subsequently, the samples were stuck to metallic supports and metallized with gold (Blatec Sputter, CPD 030). After metallization, the samples were analysed with a scanning electron microscope JEOL JSM-6390LV, located at the Central Laboratory of Electronic Microscopy of the Federal University of Santa Catarina, Brazil.

2.6 | Oospore formation in *P. viticola* resistant hosts

At the end of the growing season, all resistant hosts exhibited DM symptoms and they varied according to *Rpv* gene(s) carried by the hosts, as previously observed by Zanghelini et al. (2019). Ten symptomatic leaves from each of the varieties Baron, Brönner, Cabernet Cortis, Calandro, Calardis Blanc, Felicia, Isabella, Prior, and Regent, in addition to two advanced breeding selections (ABS) Gf. 2004–043-0015 and Gf. 2004–043-0024 were collected, fixed, and cleared. From each leaf, a 12 mm disc was excised from a region showing DM lesions, totalling 10 leaf discs per genotype. For the identification of *P. viticola* oospores, the same methodology described for the susceptible host variety, Cabernet Sauvignon, was used. Ten oospores per leaf disc were measured, totalling 100 oospores per genotype.

2.7 | Statistical analysis

The diameters of *P. viticola* oospores (μm) measured in the variety Cabernet Sauvignon sampled from Curitibanos and SJ were compared by Kruskal–Wallis test using R software (R Core Team, 2018). The same analysis was applied to compare oospore sizes from the resistant hosts. For the analysis, leaves were considered as replications.

3 | RESULTS

3.1 | Climatic conditions

During the annual grapevine cycle (September to March) the average temperature and the accumulated precipitation in Curitibanos (cycle 2016/2017) and SJ (cycle 2017/2018) were 18.4 and 16.1 °C, and 875 and 1,057 mm, respectively (Figure 1). In 2017 a reduction of 5 °C in the average temperature was observed from January (21.2 °C) to April (16.2 °C) in Curitibanos. In contrast, in 2018 only a slight temperature reduction was observed in SJ during the same months (January: 17.0 °C; April: 16.5 °C), but the temperature in January was lower when compared to previous year in Curitibanos. In Curitibanos and SJ, 640.6 and 256.4 degree days under the temperature of 14 °C, which is considered the optimal temperature for oospore formation (Gehmann et al., 1987), were accumulated respectively from January until the date of sample collection. During the 30 days before sample collection, the mean temperature was 16.9 °C in Curitibanos ranging from 4.5 to 26.3 °C, and 16.4 °C in SJ, ranging from 9.3 to 25.9 °C.

3.2 | *P. viticola* oospore formation in the susceptible host

Oospores of *P. viticola* were observed in all leaves collected from the susceptible variety Caberbet Sauvignon in Curitibanos and SJ. Oospores presented a thick wall and a diameter ranging from 20.00 to 49.15 μm, consistent with the morphology described in a previous investigation (Vercesi et al., 1999). The average diameter of the oospores measured from Curitibanos was 31.61 μm, ranging from 20.06 to 49.15 μm. The oospores from SJ were significantly smaller ($p = 6.3e{-}08$), with a mean diameter of 22.20 μm, ranging from 20.00

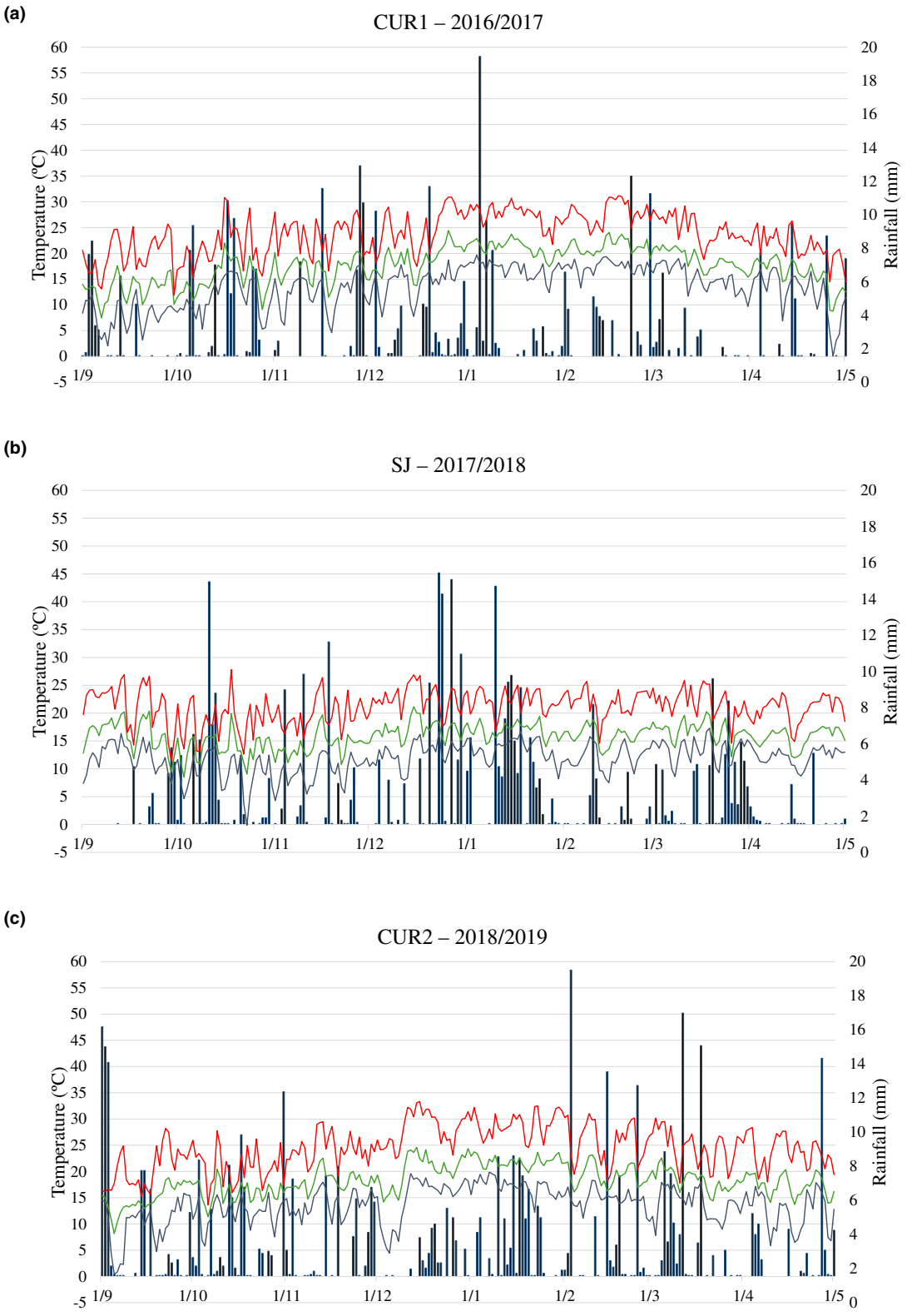


FIGURE 1 Daily (black histogram) and minimum (black line) rainfall, and average (green line) and maximum (red line) air temperatures recorded from September to May in the grapevine growing seasons of (a) 2016/2017 in Curitibanos, Brazil (CUR1), (b) 2017/2018 in São Joaquim, Bazil (SJ), and (c) 2018/2019 in Curitibanos (CUR2)

to 31.04 μm. The diameter distribution of the 600 oospores measured is shown in Figure 2. Oospores were found in all leaf discs analysed, with greater abundance in the regions close to the leaf veins and the central regions of DM lesions (Figure 3). The leaf regions rich in oospores manifested a light brown colour and sporulation (sporangiophores and sporangia) of the pathogen was rarely observed.

3.3 | Distribution of *P. viticola* oospores inside leaf tissue

Plasmopara viticola oospores were observed in the cross-sections of the variety Cabernet Sauvignon stained with eosin Y and toluidine blue dyes. The oospores were located in the central part of the leaf mesophyll, between the palisade parenchyma and the spongy parenchyma. The size and shape of the structures observed were characteristic for *P. viticola* oospores. The thick wall can be seen in more detail in Figure 4, where it is possible to observe hyphae of the pathogen penetrating the mesophyll tissue.

3.4 | Scanning electron microscopy

The scanning electron microscopy (SEM) allowed a more detailed observation of the morphology of the *P. viticola* oospore (Figure 5). The wall of the oospore is divided into two main layers, a thin outer layer and a thick inner layer. The oospore also has a layer of lipid globules. A thin plasma membrane is visible between the lipid layer and the thick inner layer, surrounding the entire oosphere. Finally, in the central region of the oospore is the ooplast.

3.5 | Formation of *P. viticola* oospores in resistant hosts

Among the resistant hosts analysed, oospores were observed in Baron, Brönner, Cabernet Cortis, Calandro, Felicia, Prior, and in the two ABS Gf. 2004–043-0015 and Gf. 2004–043-0024. No oospores were found in the hosts Calardis Blanc, Isabella, and Regent although a more detailed analysis is required to confirm this information. The regions of the discs with the highest concentration of oospores were similar to those seen in the susceptible host. Oospores were concentrated close to the leaf veins and found mainly in the central regions of DM lesions. Oospores of *P. viticola* found in a resistant host can be seen in Figure 6. Figure 6b shows hyphae of the DM between the oospore and the stoma. The diameter of the oospores found in resistant hosts ranged from 16.3 to 44.9 μm. No significant effect of resistant host variety on oospore diameter was observed ($p = 0.29$). The diameter distributions of the 100 oospores per resistant host are shown in Figure 7.

4 | DISCUSSION

The present investigation is the first to scientifically prove oospore formation in *P. viticola* occurring on leaves of grapevines in the highlands of the State of Santa Catarina, southern Brazil. As *P. viticola* is a heterothallic oomycete, and no homothallic isolates were found in previous investigations in North America (Wong et al., 2001) and Europe (Scherer & Gisi, 2006), the results suggest that the compatible mating types P1 and P2 are present in the pathogen populations studied. However, further investigations are needed to confirm this hypothesis.

Climatic conditions play an important role in the reproductive cycle of *P. viticola*. The formation of oospores is commonly associated with a physiological inhibition of asexual reproduction, occurring mainly in leaves aged in the autumn period (Grünzel, 1961). In the highlands of Santa Catarina from late summer and the beginning of autumn, a progressive decrease in temperatures and precipitation occurs, inducing the senescence of leaves and the dormancy of vines. The complete lack of green leaves causes a bottleneck for asexual reproduction of *P. viticola*, promoting sexual reproduction. In our study, the leaves with oospores collected were yellowish in colour and had mosaic symptoms caused by advanced DM infection.

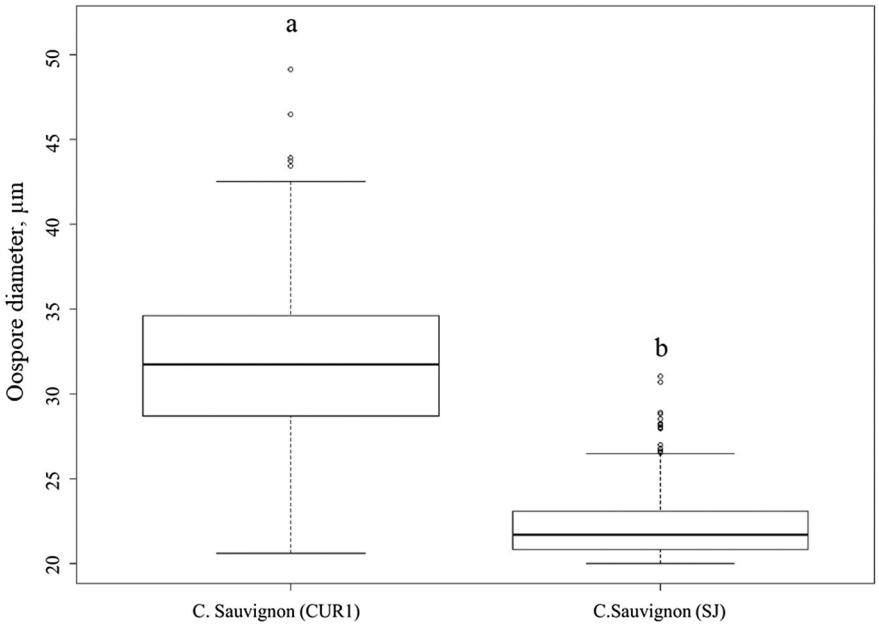


FIGURE 2 Distribution of oospore diameters (μm) of 600 *Plasmopara viticola* oospores found in the susceptible grapevine host Cabernet Sauvignon in Curitibanos (CUR1) and São Joaquim (SJ), in the highlands of Santa Catarina State, southern Brazil. The letters (a) and (b) indicate a significant difference in mean oospore diameter between the locations, detected by Kruskal–Wallis test ($p < 0.05$)

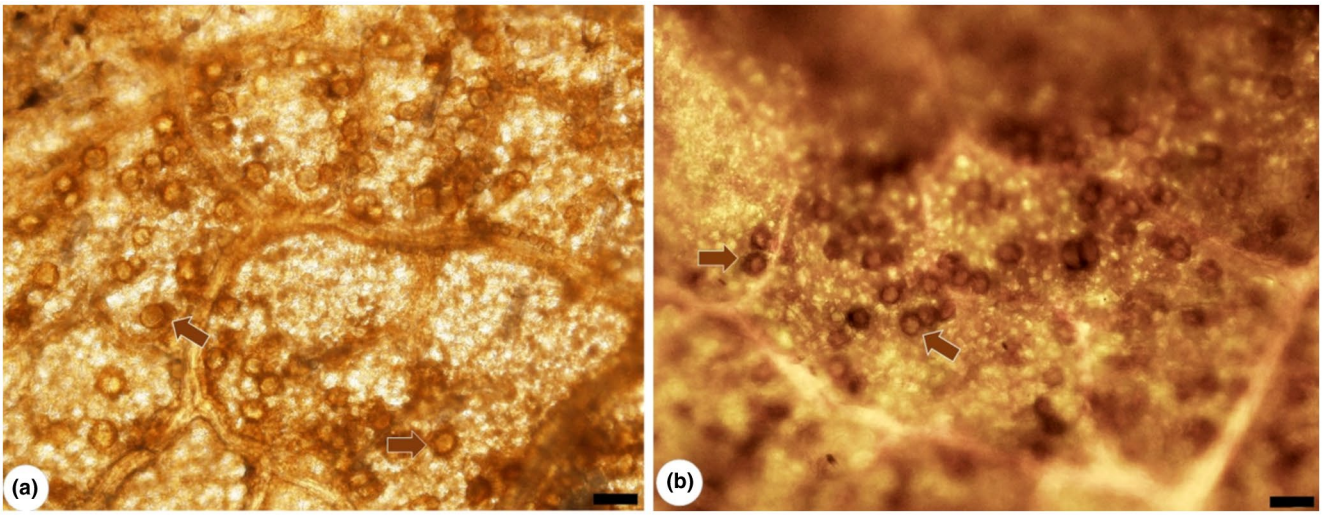
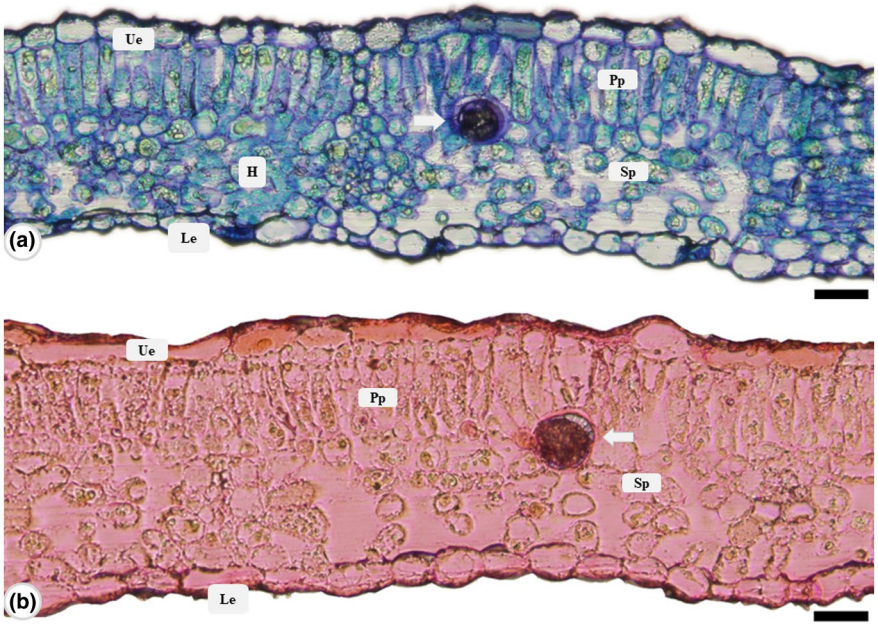


FIGURE 3 Oospores of *Plasmopara viticola* in leaf discs of the susceptible grapevine host Cabernet Sauvignon grown in highland regions of southern Brazil. (a) Central region of the downy mildew lesion with oospores (arrows) in leaf sampled in Curitibanos (CUR1) (scale = 20 μm). (b) Leaf disc rich in oospores (arrows), sampled in São Joaquim (SJ). Scale bar = 10 μm)

FIGURE 4 Cross-sections of leaves of the grapevine host Cabernet Sauvignon showing the presence of oospores. (a) Cross section of a leaf collected in Curitibanos, Brazil (CUR1), stained with toluidine blue. (b) Cross section of a leaf collected in São Joaquim (SJ), stained with eosin Y. Pp, palisade parenchyma; Ue, upper epidermis. Arrow: oospore. H, hypha; Le, lower epidermis; Sp, spongy parenchyma. Scale bar = 20 μm



In addition, asexual sporulation was no longer visible externally in lesions, characteristics also reported by Vercesi et al. (1999).

Temperature seems to play a major role in oospore formation. Although laboratory assays demonstrated that oospores were formed at a wide range of temperatures, varying from 10 to 25 °C, a higher concentration of oospores was observed at temperatures ranging from 12 to 18 °C (Gehmann et al., 1987; Lehoczky, 1956; Wong et al., 2001). In our work, at the beginning of autumn, during the period of 30 days before sample collection, the average temperature was 16.9 °C in Curitibanos, varying from 4.5 to 26.3 °C, and 16.4 °C in SJ, ranging from 9.3 to 25.9 °C, which was within the temperature range reported as the best for formation of *P. viticola* oospores.

The oospores found in the samples were concentrated in the central regions of the DM lesions and close to leaf veins. These results are consistent with the observations of Lehoczky (1956) and Wong et al. (2001). For all the studied samples, the sexual spores had a rounded shape, yellowish colour, and a thick wall, typical of *P. viticola* oospore morphology (Vercesi et al., 1999). Oospore diameters varied from 16.3 to 44.9 μm in the susceptible and the resistant hosts, in agreement with the diameter range reported in the literature (20.0–50.0 μm; Lehoczky, 1956). The significantly smaller size of the oospores in Cabernet Sauvignon from SJ compared to those in Curitibanos may have reflected different states of maturation of the oospores due to the lower GDD accumulated

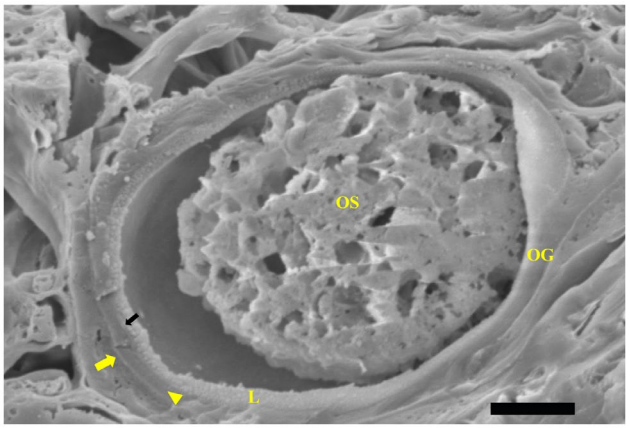


FIGURE 5 Oospore of *Plasmopara viticola* observed in the susceptible host Cabernet Sauvignon and visualized with scanning electron microscopy. Yellow arrowhead, thick inner layer of oospore wall; L, lipid globules; OG, oogonium wall; OS, ooplast; black arrow, plasma membrane; yellow arrow, outer oospore wall. Scale bar = 5 μm

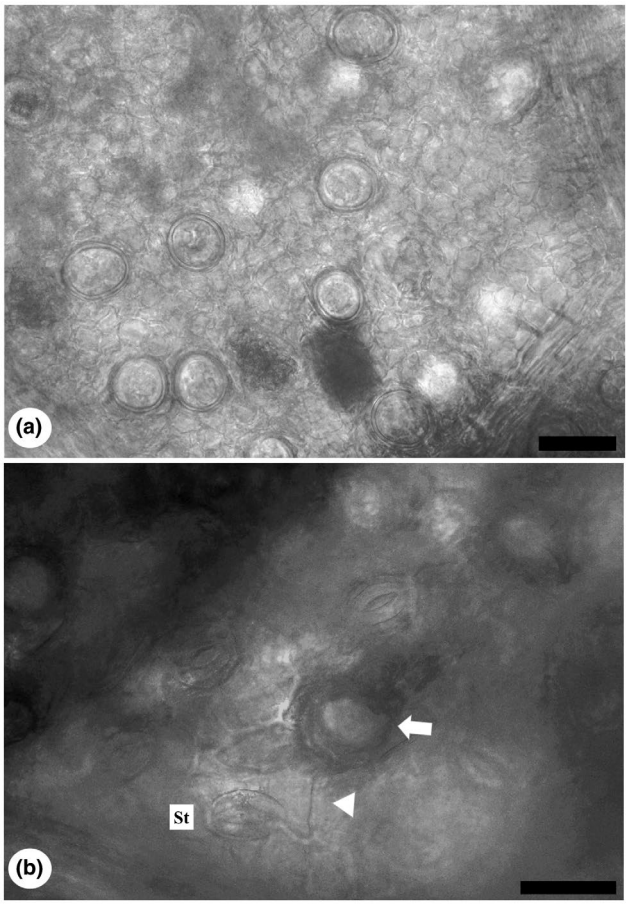


FIGURE 6 Oospores of *Plasmopara viticola* found in leaf discs from resistant grapevine host grown in highland regions of southern Brazil (CUR2). (a) Oospores in leaf discs of cultivar Brönner (bar = 20 μm). (b) Hyphae of *P. viticola* between the oospore and the stoma. St, stoma; Arrowhead: hypha; Arrow: oospore. Scale bar = 50 μm

in SJ. Similarly, Vercesi et al. (1999) identified three types of *P. viticola* oospores: type I were considered immature and are characterized by a thin internal wall, large lipid globules, and two nuclei; type II were considered mature, with a noticeable increase in the thickness of the internal wall, fusion of nuclei, and the breakdown of large lipid globules; and type III oospores have very large lipid globules, which occupy practically the entire cytoplasm region and the oospore has a very thick and quite homogeneous inner layer. The latter are considered abnormal and hardly viable (Dick, 1995). Further investigations evaluating the evolution of oospore maturation under the contrasting temperatures of southern Brazil and with different hosts need to be carried out to confirm this hypothesis.

The cross-sections of leaf tissues revealed that the oospores are in the central part of the mesophyll (between the palisade parenchyma and spongy parenchyma). Leaf mesophyll is the most active photosynthetic tissue, containing many chloroplasts and photosynthetic pigments. The most compacted regions of the tissue (palisade parenchyma) did not show the formation of oospores. The more grouped cells of plant tissue form a natural structural barrier, preventing further development of the pathogen during the asexual phase (Nascimento-Gavioli et al., 2019), and consequently, the formation of oospores. Although various studies have employed histopathological analysis to describe the formation and distribution of asexual structures of the pathogen in plant tissues (e.g., Nascimento-Gavioli et al., 2019), only a few are dedicated to the sexual reproduction of *P. viticola* (e.g., Kortekamp, 2005).

The structures that characterize the oospore morphologically were observed in SEM analysis. During the process of oospore formation, protoplasmic reorganization occurs. This reorganization involves a redistribution of three main types of reserves: the inner oospore wall, the ooplast, and lipid globules. According to Dick (1995) and Vercesi et al. (1999), the inner oospore wall is composed of β-glucans and is considered the most important layer of the structure, as it is reabsorbed in the germination process. Its volume and thickness gradually decrease before the germ tube is formed. Another important structure is the ooplast, which also contains β-glucans, but its main function is phosphate storage. Finally, the L layer is made up of lipids and is considered an endogenous energy source. Other structures that have been observed in the oospore by SEM are the outer oospore wall, plasma membrane, and the oogonium wall. All the structures observed in SEM are consistent with the description made for oospores from *P. viticola* by Vercesi et al. (1999). Oospores from *P. viticola* were also observed by SEM in the susceptible varieties Kerner and Riesling (Kortekamp et al., 1998); however, the oospores detected had a smaller diameter than those in the present study.

The identification of oospores is strong evidence of sexual reproduction. The heterothallic nature of *P. viticola* requires the outcrossing of two sexually compatible types (female and male). Diploid nuclei within both female (oogonium) and male (antheridium) gametes undergo meiosis, promoting the recombination of the chromosomes

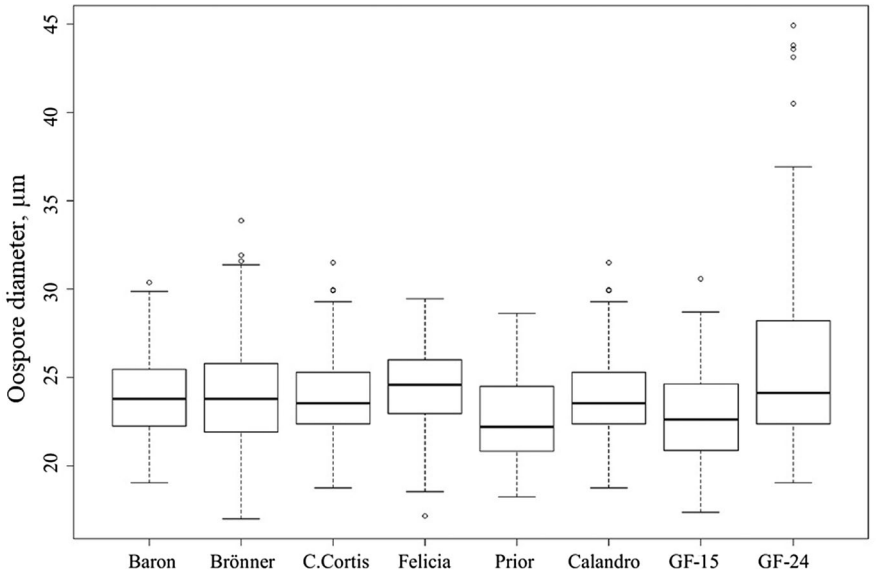


FIGURE 7 Distribution of the diameters of the *Plasmopara viticola* oospores ($n = 100$) found in resistant grapevine host cultivars Baron, Bronner, Cabernet Cortis (C. Cortis), Felicia, Prior, Calandro, Gf. 2004–043–0015 (GF-15) and Gf. 2004–043–0024 (GF-24) grown in Curitibanos, Brazil (CUR2). No significant differences were observed in oospore diameters between the different resistant hosts ($p > 0.05$)

and new allele arrangement (Burruano, 2000). Therefore, it is expected that *P. viticola* populations in the highlands of Santa Catarina are highly diverse, having direct implications on pathogen management. This is in accordance with Santos et al. (2020), who observed greater genetic diversity in *P. viticola* populations from the state of Rio Grande do Sul, southern Brazil, than in populations of the State of São Paulo, south-eastern Brazil. The populations from Rio Grande do Sul presented a higher number of distinct MLGs, a CF under 0.09, and genotypic evenness greater than 0.91, strongly suggesting sexual reproduction. In contrast, the populations from São Paulo showed lower MLGs, CF from 0.70 and genotypic evenness less than 0.56, indicating a predominance of asexual reproduction. The lower temperature during autumn and winter in southern Brazil probably plays a decisive role on the mode of reproduction of *P. viticola*.

Oospores play a key role in the epidemics of DM. Until recently, it was assumed that oospores played a major role only in primary infections at the start of the vegetative growth season of the grapevine and that progress of the disease was achieved by various cycles of asexual reproduction. More recently, studies carried out in different countries of Europe have shown that the oospores are the largest source of inoculum throughout the vegetative cycle of the grapevine (Gobbin et al., 2006). As oospore formation contributes to the increase in genetic variability of the pathogen, this finding may explain the rapid advance of the disease in the vineyards of southern Brazil. High genetic variability on a small spatial and temporal scale may provide the pathogen with rapid evolution and adaptation to different environments and hosts. The greater the diversity of the pathogen population, the greater the chances of it breaking the host's genetic resistance or developing resistance to fungicides applied for its control (McDonald & Linde, 2002).

In our work, *P. viticola* oospores were also found in PIWI varieties and ABS containing different loci of resistance, singly or in combinations, such as *Rpv3.1*, *Rpv1* + *Rpv3.1*, *Rpv3.1* + *Rpv3.2*, and *Rpv3.3* + *Rpv10*, reinforcing concerns about the durability of the resistance. The formation of viable and fertile oospores in varieties partially resistant to disease (*Rpv1*, *Rpv3*, and *Rpv1*/*Rpv3*) was also observed by Delbac et al. (2019). In addition, breakdown in resistance was reported in Europe, for example, for resistance locus *Rpv3.1* in the variety Bianca (Peressotti et al., 2010); oospores were not observed for the varieties Regent (*Rpv3.1*), Calardis Blanc (*Rpv3.1* + *Rpv3.2*), or Isabella (*V. labrusca* interspecific hybrid). However, information about the direct effect of resistance genes on the ontogeny of *P. viticola* oospores is lacking and further investigation is needed.

Our results demonstrate that *P. viticola* completes its life cycle (asexual and sexual) in susceptible varieties, PIWI varieties, and ABS in highland regions of southern Brazil. This study is of great relevance for Brazilian viticulture, being the first scientific study to confirm the formation of *P. viticola* oospores in Brazil. The occurrence of sexual reproduction has great implications in the management of the disease and in grapevine breeding because it contributes to the development of populations with high genetic variability, as recently reported by Santos et al. (2020). Such populations have a greater capacity to overcome resistance genes or develop resistance to fungicides (McDonald & Linde, 2002), challenging the efforts made to control it. During the production of *V. vinifera* grapes, numerous fungicide treatments are carried out to prevent serious epidemics of DM. In addition, it is already known that *P. viticola* is a high-risk pathogen for the development of resistance to fungicides (Santos et al., 2020; Toffolatti et al., 2010). Therefore, to avoid high selection pressure on populations and to preserve the effectiveness of resistant varieties and the products applied for their control, it is necessary to consider the formation of oospores as an important factor in the pathogen cycle and in the environment.

In-depth knowledge of the mechanisms of oospore germination can lead to new combined strategies for the management

of DM. Studies of disease prediction models should consider the climatic conditions favourable for germination of *P. viticola* oospores, because they are the primary inoculum sources throughout the vegetative cycle of grapevine (Gobbin et al., 2006). Our finding of oospores in grapevines of southern Brazil may contribute to the accuracy of prediction models, and they, in turn, can improve the efficiency of disease control. In addition, studies should be carried out to evaluate the elimination of the main source of inoculum at the end of the growing season, as this may reduce the number of oospores present in the soil and consequently reduce infection over the years. It is a common practice to reduce or eliminate phytosanitary management after grape harvest; however, this allows a window for DM infection at the end of the vegetative cycle, which, according to our study, will lead to the formation of oospores. Eliminating DM on vegetation at the end of the growing seasons should have significant impact on the genetic diversity of *P. viticola* populations in the following seasons. In addition, advanced studies to evaluate the biology and genetic structure of *P. viticola* populations in this and other regions are necessary. Such studies would allow the determination of the impact of oospore formation on the genetic structure of *P. viticola* populations in southern Brazil to be validated and contribute to the formulation of strategies to improve the efficiency of DM management. This knowledge is also essential to define grape breeding strategies for lasting resistance to DM, the main grapevine disease in southern Brazil.

ACKNOWLEDGEMENTS

We are thankful for the UNIEDU/FUMDES programme in the state of Santa Catarina for granting a scholarship to C.B. We would also like to thank the technicians of the Central Laboratory of Electronic Microscopy (LCME) of the Federal University of Santa Catarina for their assistance in SEM.

CONFLICT OF INTEREST

The authors have no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Bitencourt C, Pierre PMO, Pinto FAMF, et al. First report of oospore formation in *Plasmopara viticola*, the causal agent of grapevine downy mildew, in highland regions of southern Brazil. *Plant Pathol*. 2021;00:1–11. https://doi.org/10.1111/ppa.13431